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The Blockchain and Tokenization for Social Good: Empowering Smallholder Farmers in Zimbabwe's Agricultural Supply Chains

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Abstract- Smallholder farmers are the cornerstone of global agricultural value chains but are systematically excluded from the formal financial system, subjected to crippling obstacles such as unavailability of collateral and obscure patterns of trading. This situation is practically acute in Zimbabwe, where farmers face devastation due to crippling financial exclusion fueled by hyperinflation, insufficient collateral, and fragmented credit markets. This study examines how tokenization and blockchain can address this exclusion. The research is informed by Institutional Theory and the Capabilities Approach. A qualitative single-case study utilizing purposive sampling enrolled 32 participants from Zimbabwe's agricultural sector. Data were collected through semi-structured interviews (n=25) between agri-tech innovators, smallholder farmers, and union representatives, augmented by four focus groups and documentary analysis. Data were analyzed thematically using NVivo Version 15 with trustworthiness ensured via triangulation and member-checking. The results suggest that tokenizing farm assets has the potential to make pre-harvest finance accessible and bring transparency. Major impediments like digital literacy deficits, infrastructural constraints, and regulatory ambiguity inhibit adoption. The study concludes that blockchain must be co-designed within a zone which ranges from institutional deficits to farmers' financial capabilities and steers clear of technical solutionism if it is to unlock its transformative potential.

Keywords: Blockchain; Tokenization; Smallholder Farmers Zimbabwe; Financial Inclusion; Supply Chain Finance.

1. INTRODUCTION

Financial exclusion remains a formidable global challenge, especially among smallholder farmers who comprise a large percentage of rural populations in Sub-Saharan Africa. After decades of policy attempts, the majority of them remain incapable of accessing formal credit, savings, insurance, or other financial services at terms allowing them to invest, grow, or buffer shocks. Researchers like Adegbite and Machethe (2022) have found that in Nigeria, financial inclusion reduces farm income reliance by 37.6% and increases non-farm economic activity by around 32.5%, but still, a considerable percentage of smallholder farmers are excluded because of multidimensional challenges of geography, financial literacy, and institutional trust. In Zimbabwe, formal adult financial inclusion rose from 69% in 2014 to 84% in 2022, while exclusion fell from 23% to 12%, (FinMark Trust, 2022). Rural populations, smallholder farmers, women, and youth lag behind in the uptake of second-generation products with features beyond a simple transaction account (FinMark Trust, 2022). Zimbabwe's agricultural sector has suffered from economic volatility, hyperinflation, erratic monetary policy, and the collapse or retrenchment of traditional agricultural finance channels.

Land reform changes and insecure tenure, weak rural banking infrastructure, and high transaction and collateral costs further compound challenges for smallholder farmers. Masuka et al. (2025) show that although 94.4% of smallholder farmers in sampled areas achieve basic inclusion (i.e., having access to at least one or more formal financial instruments), only 25.6% receive various financial services due to a myriad of challenges. In the bigger scheme of things, limited education, large family sizes, gender gaps, and high transaction costs driven by underdeveloped rural banking characterize the rather inclement environment (Masuka, 2025).

In this strained economy, ideas around tokenization and blockchain propose some transformative potential. Indeed, a few researchers have started to explore how tokenizing farmland with smart contracts and using DeFi could narrow information gaps, enhance transparency, accelerate loan disbursement, and reduce costs. As an example, Olateju (2025) finds that tokenization shortens loan disbursement times by ~31.3%, reduces transaction costs by ~37.3%, and enhances financial transparency by ~40%, together with a ~15% reduction in fraud in operations with blockchain. Kumarathunga et al. (2022) developed the idea of a Smart Agricultural Futures Market where smallholders can trade futures contracts using blockchain to get cash in advance, consolidate marketing, and hence increase margins. Meanwhile, in South Africa, Mavilia and Pisani (2021) note through literature review that blockchain can potentially reshape agrifood value chains, improve distribution, and minimize food system inefficiencies, though they caution against the challenges of adoption, technology cost, and regulation. Despite the potential, empirical studies are few.

Several studies have been conceptual or pilot in nature. For example, The Blockchain Technology for Financial Inclusion of Smallholder Farmers in Africa (Elkoraichi et al., 2025) addresses potential, though with few case studies within weak economies or social empowerment impacts. Ugandan studies of barley farmers (Ninsiima et al., 2025) examine intention to use blockchain using the Technology Acceptance Model (TAM), showing that trust, perceived ease of use, and perceived usefulness are significant; yet the sample is rather localized, the adoption level remains low, and downstream

implications (for credit access, social influence, and supply chain trust) remain limited in terms of detailed investigation (Muzondo et al., 2025).

In such an exposed economic condition, the introduction of blockchain and tokenization technologies brings in potentially disruptive innovations. Scholars have begun exploring how the tokenization of agricultural assets, smart contract usage, and decentralized finance (DeFi) structures can be used to reduce information asymmetries, improve transparency, and reduce loan disbursement times and costs.

Thus, the problem this study seeks to address is two-fold. Firstly, A finance gap still grips Zimbabwe's smallholders: countless farmers can't tap into a range of financial services, can't use formal finance to boost agricultural performance, and remain easy targets for poverty traps. Second, there is a research gap: few studies have empirically examined the implementation of blockchain/tokenization in fragile or volatile economies (especially Zimbabwe), and there is a lack of rigorous data on operational barriers, social dynamics (trust, power relations), and measurable social empowerment outcomes.

This article bridges such gaps by asking: How can blockchain and tokenization be leveraged to promote financial inclusion and social empowerment of Zimbabwean smallholder farmers? Sub-questions investigate the tokenization of agricultural assets, its process and benefits; establish operational, technical and social challenges in the Zimbabwean context; and examine how these technologies affect power structures and trust along agricultural supply chains. The urgency of this inquiry is accentuated by the poor performance of the Zimbabwean economy, the recent shocks to inflation, and the need to revitalize agriculture for food security, foreign exchange earnings, and rural incomes. Without thorough, localized research, policy and action will likely fail, become misallocated, or increase inequalities.

Through empirical field research, analysis of tokenization model, and stakeholder interviews, this study aims to make a theoretical contribution (extending knowledge of trust, finance, and technology adoption in fragile states), a policy contribution (regulation, infrastructure, and inclusion advice), and practical contribution (tokenization tool development appropriate for Zimbabwe). The anticipated significance is to offer pathways for smallholder farmers to secure not only financial inclusion but also robust social empowerment through inclusion, agency, trust, and sustainable agricultural transformation.

2. THEORETICAL FRAMEWORK

2.1 Theoretical Framework

To provide context for the study, Institutional Theory and the Capabilities Approach are complementary explanations. Institutional Theory enables one to consider how formal institutions (law, regulation, contracts) and informal institutions (trust, norms, networks) shape new technology and financial product adoption (Boakye et al., 2023; Mthimkhulu & Jokonya, 2022).

The Capabilities Approach (Sen, 1999; Nussbaum, 2011; Robeyns, 2017) looks beyond what farmers have (access, resources), but also beyond what they are able to do and be; their real freedom, agency, and capability to convert access into outcomes. Synthesizing the theories enables a two-stage analytic framework: one, to chart institutional enablers and barriers; two, to measure whether blockchain or tokenization adoption results in enhanced functioning (for example, enhanced negotiation power,

planning capacity, and risk management). This synthesis is particularly important in Zimbabwe, where institutions are precarious, regulatory regimes shift, informal norms are robust, and numerous smallholders lack conversion capacities (digital literacy, infrastructure, and trust).

2.2 The State of Smallholder Agriculture and Finance in Zimbabwe

Existing statistics show that while most Zimbabwean smallholders are included to some degree, higher inclusion is limited. Masuka et al. (2025) note that 94.4% of smallholder farmers have basic access to one or more formal financial services, yet only 25.6% utilize multiple services (credit, savings, insurance, etc.). In addition, Kawara et al. (2025) also discover heartening developments, yet massive figures are still with informal mechanisms, as commercial bank policies are not accommodating enough to rural conditions. In Chiredzi, resettled sugarcane farmers' size and need for microcredit are constrained by risk and long maturities, such that even if credit is accessible, smallholders will tend to suffer from timing and size issues (Matsvai, 2024). Agricultural growth is said to be positively related with bank credit in long-term terms, but the short-term effects are uncertain due to the external shocks, policy uncertainty, or uncertainty in climatic conditions (Chikwira, 2024).

These studies point to systemic impediments: excessive transaction costs, primitive rural infrastructure, education and gender gaps, weak financial literacy, regulatory and market risks. The evidence says access is not merely adequate but essential: quality, terms, diversity, and timing of the finance are of concern.

2.3 Blockchain and Tokenization in Supply Chain Finance (SCF): A Global Perspective

Blockchain and tokenization have strengthened SCF technologies globally. Boakye et al. (2023) investigate blockchain SCF adoption of SMEs in Ghana. These authors find that the technical dimensions-including relative advantage, cost, and complexity are the determinants of blockchain SCF adoption, while environmental pressure, which involves competition and changing market dynamics, plays less of a role. Ioannou and Demirel (2022) illustrate how blockchain can minimize inefficiency, increase visibility, allow for much higher trust between supply chain partners, and give way to the intersection of law, finance, and operations. In China, Li et al. (2024) show that firms adopting blockchain face lower financing constraints in data from 2016-2021, especially when firm size is moderate, regulatory support exists, and environmental pressures are favorable. Also, in South Africa, Mhlanga and Shao (2025) demonstrate how blockchain contributes to sustainable supply chain finance, increasing profitability and environmental responsibility while also encountering regulatory and infrastructure barriers.

These studies provide useful mechanisms: traceability, smart contracts, transparency, automated verification, and reduction in intermediaries. However, few of these studies focus on tokenization of agricultural assets (e.g. tokenizing crops, future harvest, and land) in Zimbabwe or similar fragile economies; many focus on SMEs more broadly or use blockchain in operational or logistic settings rather than in financing smallholder agriculture.

2.4 The Intersection of FinTech, Agriculture, and Social Impact in Developing Economies

There has been a notable increase in research examining the effects of digital finance on marginalized communities and smallholders. For example, Mutale and Shumba (2024) show that mobile money and digital banking in Zimbabwe have expanded financial services access and improved financial literacy among marginalized groups through cost savings, faster transactions, and enhanced security. In rural Zambia, Mwalupaso et al. (2025) find that financial inclusion via village savings and loan associations (VSLAs) significantly increases adoption of climate-smart agriculture among women, with digital agricultural tools mediating much of this effect. The global review by Mapanje et al. (2023) find that digital finance helps with procurement of inputs, market access, and risk spreading but is mostly delivered via loans, mobile money, or payment systems, not via tokenization of assets. Likewise, the viewpoint of Asante et al. (2023) support that the aspects of relative advantage, cost, and complexity play important roles in the adoption of blockchain supply chain finance for Ghana's SMEs, which indicates that although there is consideration, adoption is low.

Moreover, there is evidence that social outcomes, such as empowerment, gender equity, or trust, have been assessed rarely (Malapit & Quisumbing, 2015). Often, researchers use cross-section analyses, or assess intentions, as opposed to realized outcomes (Malapit & Quisumbing, 2015). The importance of supply chain power, the nature of trusts in relationships between farmers or producers, or whether technology drives inequity has not been analyzed sufficiently.

2.5 Identified Knowledge Gaps

Several gaps emerge clearly. Firstly, in Zimbabwe's smallholder agriculture, little empirical work has been done on blockchain and tokenization, particularly tokenizing agricultural assets. Secondly, most existing studies focus on access or intention rather than outcome measures like changes in income, agency, trust, or power dynamics (Lim & Meer, 2016; Joshi & Houtzager, 2021). In addition to that, there is limited work combining institutional analysis (formal/informal institutions) with capability outcomes (freedoms, agency, conversion factors). Furthermore, quantitative data on cost savings, transaction time, trust gains, and risk reduction due to blockchain/tokenization under fragile economy conditions are sparse. Finally, while many barriers are identified (regulation, infrastructure, cost, literacy), there is insufficient clarity on which ones are the most binding in Zimbabwe and how they interact, or how they might be overcome in locally feasible ways (Muzondo et al., 2025).

3. METHODOLOGY

This study adopted an interpretivist paradigm to explore the meanings, experiences, and social dynamics surrounding blockchain/tokenization in smallholder finance in Zimbabwe. In line with interpretivism, the goal was not to test hypotheses but to understand the phenomena from the perspectives of the actors themselves, recognizing that reality is socially constructed (Schwandt, 2014; Creswell & Poth, 2018). To this end, a qualitative single-case study design was selected: the case being the entire implementation ecosystem of blockchain/tokenization projects for smallholder finance in Zimbabwe (Yin, 2018). A case study design was appropriate because it allowed in-depth, contextualized exploration of a bounded phenomenon (Stake, 1995; Yin, 2018), especially where boundaries between

phenomenon and context are fluid and where multiple stakeholders interact (Baxter & Jack, 2008). Similar qualitative case studies have been used in fintech, agri-tech and supply chain research to unpack complex socio-technical processes (e.g., McLeod & Dulsky, 2021; O'Sullivan et al., 2022).

The single-case design sufficed since it enables an in-depth investigation into the contextualized nascent blockchain ecosystem in Zimbabwe, bound by the fact that the context was important in understanding implementation barriers and social dynamics (Yin, 2018). While such a limited context does not offer broad statistical generalization, this study presents analytical generalization, opening a window into institutional and socio-technical processes which may also apply to comparable fragile economies. The main limitation consists in the contingency of specific findings on the unique Zimbabwean regulatory, infrastructural, and economic context.

The population comprised all stakeholder groups actively involved in agricultural finance, technology development, regulation, and implementation in Zimbabwe; this included smallholder farmers, farmer union leaders, agri-tech startup owners or managers, fintech experts, government/regulatory bodies, and development partner/Non-Governmental Organization (NGO) actors engaged in agricultural finance or technology initiatives.

Given the heterogeneity of stakeholder roles and the need to deeply capture voices across groups, the study adopted purposive sampling to deliberately select participants who hold rich, relevant knowledge of blockchain/tokenization in agricultural finance. Purposive sampling is widely recommended in qualitative research to maximize information richness and relevance (Patton, 2015; Miles et al., 2020). Inclusion criteria included stakeholders actively working in the agricultural finance or technology sector in Zimbabwe, which have direct experience or decision-making authority in the relevant domain of smallholder finance or blockchain innovation.

Exclusion criteria filtered out potential participants with any other type of role or knowledge not relevant to the phenomena under study. Participants were accessed through the professional networks of agri-tech incubators and farmer unions, with snowball sampling toward specific niche experts (Noy, 2008). The sample size of 32 participants was determined through thematic saturation. Subsequent interviews and focus groups did not provide new substantive themes relevant to the research questions, hence confirming informational redundancy across key stakeholder categories (Guest et al., 2006; Mason, 2010). 15 small-scale farmers from various provinces (Mashonaland, Manicaland, and Matabeleland) were selected to ensure geographic variability; 4 leaders of farmer unions (e.g., from Zimbabwe Farmers Union branches); 5 founders/managers of agri-tech startups; 3 FinTech specialists; 3 representatives of government/regulatory agencies (e.g., central bank, fintech regulatory office, and ministry of agriculture); and 2 representatives of development partners/NGOs engaged in Agri finance or rural tech initiatives.

In data collection, primary data were gathered through semi-structured interviews and focus group discussions (FGDs). A total of 25 individual in-depth interviews (each taking 45–60 minutes on average) were conducted with all the stakeholder groups, with a flexible interview guide allowing probing of emergent issues. Semi-structured interviews are best suited to learn how individuals make sense of what is happening to them and to balance structure with openness (Kallio et al., 2016; Brinkmann & Kvale, 2015). Furthermore, four FGDs were organized with smallholder farmers (5–7 farmers per group)

to obtain collective accounts, rank order typical issues, and make discussion of blockchain/tokenization in a setting feasible (Morgan, 1997; Krueger & Casey, 2015). FGDs cross-checked interview data and enriched contextual data. Besides primary data, secondary data included document analysis of project reports, whitepapers of blockchain platforms, company websites, and national policy/strategy papers regarding fintech, agriculture, and rural finance.

After data collection, all interviews and FGDs were tape-recorded (with consent) and verbatim transcribed. The transcripts were then exported and managed using NVivo software (Jackson & Bazeley, 2019; Allsop et al., 2022) to facilitate systematic coding, questioning, and theme development. The thematic analysis followed Braun and Clarke (2006)'s six-stage process with minor modifications in line with more recent developments: (1) familiarization (reading and re-reading transcripts), (2) generating initial codes, (3) searching for themes (collating codes into candidate themes), (4) reviewing themes (improving coherence and consistency with data), (5) defining and naming themes (producing clear theme definitions), and (6) producing the report (selecting illustrative quotes in relation to research questions and in relation to theory) (Christou, 2023; Braun & Clarke, 2006). NVivo's querying abilities, node hierarchies, and memos made systematic coding, cross-case comparison, and audit trails possible. Thematic analysis was chosen because of its scope for flexibility and rich, patterned meaning generation over a dataset while still remaining accessible to theory building (Kiger & Varpio, 2020).

To ensure ethical integrity, informed consent was obtained from all participants, all transcripts and identifiers were anonymized, confidentiality was assured to the participants, and the right of withdrawal at any time was explained clearly. To preserve confidentiality, full interview transcripts were not shared. However, anonymized, aggregated data summaries or thematic extracts are available from the corresponding author upon reasonable request, subject to ethical approvals. All participant identifiers were removed and replaced with non-identifiable codes (e.g., F1-F15 for farmers, U1-U4 for union representatives, T1-T5 for tech innovators, R1-R3 for regulators).

To increase trustworthiness and rigor, a number of strategies were employed. Credibility was increased through triangulation between stakeholder groups (farmers, tech actors, regulators) and between methods (interviews, FGDs, documents) (Patton, 2015; Lincoln & Guba, 1985). Member checking was undertaken by feeding back draft themes and findings to some participants to validate interpretation. Transferability was aimed for by the inclusion of a thick description of the Zimbabwean context, participant groups, and implementation settings so that readers can make judgements of applicability to other settings. Dependability was achieved through the maintenance of an audit trail, detailed logs of methodological decisions, coding memos, and version histories. Confirmability was facilitated by reflexivity: the researchers logged their positionality, assumptions, and potential biases in memos and was mindful during analysis not to project unduly (Finlay, 2002; Berger, 2015).

To mitigate selection bias, purposive sampling targeted maximum variation across stakeholder groups. Given the authors' prior advocacy for financial inclusion, researcher positionality bias was actively managed through reflexivity memos documenting assumptions, data triangulation across farmer, tech developer, and regulator perspectives. Interviewer bias was minimized in that the semi-structured guide contained open-ended questions, allowing participants to take the lead in the discussions.

4. THEMATIC ANALYSIS AND FINDINGS

4.1 Demographic and Contextual Profile of Participants

Participants were drawn from various backgrounds: smallholder farmers in remote rural villages, leaders of farmer unions, proprietors of agri-tech startups, fintech experts, regulators, and NGO/development partners' representatives. Smallholder farmers were mainly old people with limited formal education, relying mainly on traditional farming practices and subsistence agriculture. The nearer to cities, the higher the access to electricity, connectivity, and even some exposure to digital technologies; the more remote, the higher the network disruptions, power unreliability, and low digital literacy. Fintech and agri-tech respondents were more educated, younger, more mobile, and more financially innovative. Regulatory and NGO respondents offered more systematic answers, based on policy, institutional demand, and scaling issues. This difference enabled findings to include both bottom-up experiential understanding and institutional knowledge.

4.2 Presentation of Themes

Theme 1: The Mechanics of Empowerment: Tokenization as a Gateway to Collateral-Free Credit.

Most interviewees outlined tokenization as a bridge that could enable smallholders to gain credit without traditional, physical collateral. For instance, one smallholder farmer noted, "If my bags of maize or my animals can be tokenized, I don't need the land title to access a loan." A fintech expert emphasized this point by noting, "Tokens based on harvest estimates or yields lessen the need for formal property." Farm leadership representatives were more conceived: "Even if they provide tokens... banks always want something that can be repossessed, not just digital claims." Closer inspection reveals tension: tokenization has real potential to lower exclusion for landless farmers, but its effectiveness hinges on institutional acceptance. Banks, regulators, and legal frameworks must recognize tokenized assets as collateral, and at present that isn't universal.

Theme 2: The Transparency Paradox: Increased Trust vs. Perceived Loss of Autonomy.

This was summed up by one farmer: "When everyone can see the chain, I won't be cheated on payments or weight." For their part, regulators believed openness would make it easier for them to do their work: "With blockchain, we can trace movements in a way paper ledgers never allow." Yet some smallholders felt such added transparency could put pressure on them to sell their livestock more hastily-or at less favorable prices-since everyone would be watching. The bottom line is that greater transparency has the potential to enhance integrity but also opens up thorny questions about privacy, autonomy, and rights to information that demand careful balance and protection. For example, in some cases, the reason that a farmer has a large stock or high figures of production can invite coercive pricing or exploitation by the buyers.

Theme 3: The Digital Divide: Infrastructural and Literacy Barriers as Barriers to Inclusion.

Infrastructural problems and computer literacy posed as cross-cutting challenges. Rural respondents reported instances of inconsistent power supply, slow internet

connectivity, and limited devices. One farmer said, "Charging my phone takes hours when there is no power; how do I operate a digital wallet?" Another said that wonderful designs and software are all for nothing if they assume high literacy or English language capability. Leaders of the union have demonstrated that whenever technology fails (network drop, power failure), there is an instant loss of confidence. Real-world examples from Zimbabwe: the EcoFarmer platform extends far but occupies more space in towns; the Zimbabwe Digital Agriculture Assessment has claimed that rural coverage is very patchy and smartphone penetration is low.

Accordingly, the digital divide is more than a concern with regard to connectivity and technology. Rather, it includes fallback solutions, cultural practices, languages, and capabilities. Unless this is remedied, the use of tokenization will only amplify the existing inequalities.

Theme 4: Navigating the Institutional Void: Regulatory Uncertainty and the Path to Scalability.

One of the mutual observations was that a regulatory environment with ambiguity, absence of legal recognition for digital/crypto assets, policies, and frameworks hinders scale. One of the founders of a startup was quoted as saying, "No law says tokenized farm produce is legal collateral; banks reject it by saying there's no legal precedent." Regulatory respondents admitted that laws on blockchain, tokenization or crypto technologies used in agriculture lending are often addressed on an ad-hoc basis; there is no consistent, friendly policy framework. NGO/development partners have indicated that most pilots remain small and cannot scale because when the external funding or technical support is withdrawn, there is little institutional base that can hold things together. Some of the examples include unclear roles of the central bank, no laws enabling smart contracts, no dispute resolution mechanisms, and no token standardization.

Critically, scale will depend on the development of legal frameworks, institutions of trust, standardized protocols, and the confidence that regulators, banks, and farmers can appreciate and execute tokenized claims. Otherwise, tokenization efforts are likely to be stuck at pilot or niche levels.

The following table synthesizes the four primary themes identified from the thematic analysis, summarizing key findings, illustrative quotes, and their critical implications for implementation.

Table 1. Summary of Key Findings

Theme	Key Findings	Example Quotes	Implications / Critical Insights
Theme 1: The Mechanics of Empowerment	Tokenization seen as a potential path to collateral-free credit; hope exists but institutions often still demand physical collateral.	“If my maize bags or livestock could be tokenized, I would not need a piece of land title to get a loan.” “Tokens backed by harvests ... reduce the need for formal property.” “Even with tokens ... banks still want something they can seize, not just digital claims.”	Unless legal, financial, and institutional partners accept tokenized assets as collateral, empowerment remains limited. Legal reforms and lender risk assessments are required.
Theme 2: The Transparency Paradox	Transparency increases trust, and traceability; but also raises concerns about loss of autonomy, and data misuse.	“When everyone can see the chain of custody, I know I won’t be cheated ...” “With blockchain, we can audit flows ...” “If I’m too visible, ... I might be under pressure ... accept lower offers ...”	Program designs must include privacy safeguards, and control over data sharing and ensure that transparency doesn’t create external pressures harming farmers.
Theme 3: The Digital Divide	Poor infrastructure (power, connectivity), low digital literacy, and device access issues severely limit participation.	“Even charging my phone costs hours when there is no power; ... run a digital wallet?” “Excellent apps are unusable if ... assume high literacy ...” When tech fails (network/power) trust is lost.	Tokenization projects must integrate low-tech alternatives, offline functionality, training, and local languages; infrastructure investment is foundational.
Theme 4: Institutional Void & Scalability	Regulatory uncertainty, legal non-recognition, lack of standardized frameworks, dependency on pilots.	“No law says tokenized farm outputs are legal collateral ...” Policies handled case by case ... no precedent.” Many pilots remain small ... Once support ends ... little institutional backbone.	Policy makers should institute policy and regulatory frameworks that formalize, standardize and recognize tokenization as acceptable collateral to enhance smallholder farmers’ access to credit facilities.

Source: Prepared by Author

5. DISCUSSION

5.1 Interpretation of Findings

The findings indicate that tokenization can act as an institutional innovation and capability-enabling mechanism for smallholder farmers, but its effect is highly mediated by institutionally and socio-technically imposed restrictions and institutional architecture (Muzondo et al., 2025).

To the first research question: Evidence from data indicates that tokenization is involved in co-evolving informal new norms regarding what can be accepted as collateral and trust. For example, stakeholders noticed that tokenized agreements or electronic claims begin to function in the minds of lenders as collateral substitutes even without legal status. This is in line with Sanga & Aziakpono's (2025) assertion that institutional quality mediates the impact of FinTech development for digital finance for SMEs; if regulatory regimes are weak or one-sided, new institutions like tokenization could exist but with lesser effectiveness. Similarly, the e-voucher / digital input system launched by Agriculture Finance Corporation (AFC) Land and Development Bank (with Food and Agriculture Organization (FAO) under Zimbabwe Emergency Food Security Program) shows an emergent institution: a hybrid digital platform transforming norms of accountability and distribution in input provision (FAO & AFC Land and Development Bank, 2024).

To the second research question, some of the abilities are indeed increasing: finance access without collateral (for those with good token designs), transparency in transactions, and improved input access through digital means. But conversion factors (i.e., factors converting access into meaningful freedom) are feeble: digital literacy, infrastructure (electricity, connectivity), and legality is patchy. This is reflected in Uganda (Ninsiima et al., 2025), where perceived usefulness and perceived ease of use were key, but actual adoption was thwarted by norms and perceived risk.

In general, the macroeconomic environment also influences capability growth: Al-Assaf et al. (2025) illustrate that inflation, exchange-rate volatility, and price volatility reduce the magnitude of digital finance growth, which would include tokenization. Thus, tokenization is constructing new informal institutions; shared expectations around how digital assets, smart contracts, and verifiable harvests or yields can take on financial functions, and the trailing adoption of formal institutions. Capability-wise, farmers now possess new forms of agency (negotiation power, input sourcing, traceability), but freedoms in general, such as to be able to depend fully on tokenized assets in a legal dispute or to avoid extractive practices, are not made available for all.

Proposed conceptual Model Diagram

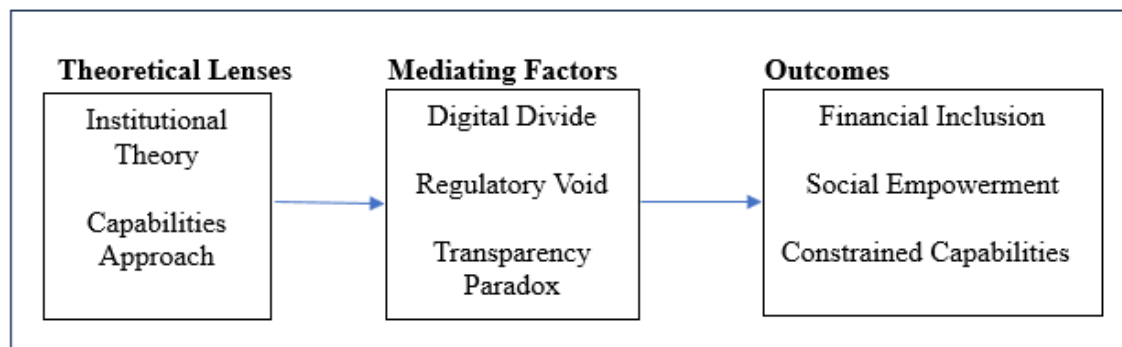


Figure 1: *Conceptual Model*

5.2 Theoretical Implications

These findings support, confirm, and expand on aspects of institutional theory and the capabilities approach. Institutionally, these findings confirm that informal institutions (norms, trust, shared expectations) generally come before formal institutions in digital innovation: tokenization probably gain traction where social norms permit trust in digital documents and where regulatory uncertainty has not choked off experimentation. This is consistent with hybrid theories of institutional change in which institutional ambidexterity (stability but innovation space) is paramount. Mweha (2025) brings to light institutional ambidexterity, a strength to sustain core institutional operations while facilitating innovation, which the Zimbabwean example establishes as required. From the perspective of the ability, the research confirms access alone is insufficient in the absence of very strong conversion drivers; this aligns with broader FinTech literature (Appiah & Agblewornu, 2025) that demonstrates legal risk, privacy concerns, literacy, and infrastructure are hindrances.

The findings discount more minimalist theories of technology adoption (e.g., TAM) in that they contend that perceived usefulness directly result in outcomes; instead, evidence shows that perceived usefulness result in higher intention, but that actual empowerment (legal enforceability, autonomy, risk management) vary institutionally. Overall, the implications are that there need to be a more cohesive theoretical framework; one that encompasses institutional change (formal/informal) and capability development as dialectical, mediated by socio-technical infrastructures, juridical institutions, and power relations. Tokenization not only reduces friction; it also re-composes what may be counted as collateral or evidence, flips power on its head, and can empower, but only if there are institutional support and social legitimation.

5.3 Practical Implications

For agri-tech startups, tokenization success in Zimbabwe (and similar contexts) depends on consideration of conversion factors: making token systems usable (low literacy, local language, offline usability), user control and data privacy, and building trust through transparency but also user control. Cooperation with local institutions (local authorities, farmer associations) supports bridging trust and legitimacy gaps. For instance, pilots for tokenization need to be carried out in co-design with farmers such that the definitions of tokenized assets (harvest, produce, future yield) are enforceable, meaningful, and transparent within some legal or customary framework.

5.4 Policy Implications

For policymakers, the evidence suggests the requirement for legal recognition of digital claims, smart contracts, and tokenized assets. Governments must consider adjustments that allow tokenized agricultural assets to be widely accepted as collateral within the law, establish regulation and standards for platforms, offer regulatory sandboxes, and facilitate infrastructural investments (power, connectivity), especially rural broadband, in order to reduce the digital divide. Additionally, e-voucher or credit card-based schemes for input distribution (like the one being piloted) show that electronic financial intervention can become more effective, transparent, and credible if corruption, go-betweens, or delay are addressed. The new farmers' credit card scheme under AFC Holdings is encouraging in this regard. The main thrust of capacity building with

development partners and NGOs encompasses digital and financial literacy, data privacy, and rights awareness while empowering farmers in these remote areas to access the required technology.

This needs to be followed by thorough testing of the financial and social effects that assess not only adoption but distributive impacts, autonomy, and power dynamics.

5.5 Recommendations

Based on the findings, and from their theoretical and practical implications, several policy suggestions arise:

1. For policy makers and regulators: the call is to provide legislation or legal provisions that recognize tokenized farming assets and digital claims as valid collateral with clarity over ownership, enforcement, and dispute resolution.
2. For banks and MFIs: Develop risk assessment frameworks for tokenized collateral, as well as pilot lending programs in conjunction with technology platforms.
3. Development agencies/NGOs: Provide funding for rural infrastructure-broadband and locally focused programming on digital and financial literacy.
4. For technology platforms: Codesign user interfaces in collaboration with farmers, focusing on low-literacy and offline functionality; have transparent data ownership rules.

6. CONCLUSION

6.1 Summary of the Study

The study explores how blockchain and tokenization are being envisioned and potentially deployed in Zimbabwe to enhance financial inclusion and empower smallholder farmers. It addresses three sub-questions: what envisioned mechanisms and benefits come with tokenizing agricultural assets, what operational, technical, and social challenges exist in the Zimbabwean context, and how these technologies may influence trust and power dynamics within the agricultural supply chain. Based on interviews, focus groups, and document analysis, the findings indicate that the majority of the stakeholders perceive tokenization as a feasible path toward collateral-free credit, increased transparency, and improved access to value chains. On the other hand, infrastructural constraints related to electricity, the internet, and access to devices, digital and financial illiteracy, regulatory uncertainty, weak legal recognition of tokenized assets, and issues of loss of control proved to be strong barriers. The capability expansion for farmers is partial: while some have new financial and informational agency, full freedoms, legal enforceability, autonomous decision-making, and scalable access are unevenly realized.

6.2 Conclusion on the Research Questions

To answer the main research question, how can blockchain and tokenization be leveraged to promote financial inclusion and social empowerment for smallholder farmers in Zimbabwe? The study concludes that they can do so but only under certain conditions.

Tokenization offers mechanisms for collateral-free or movable collateral credit, better traceability, and improved negotiation positions. Yet, unless formal institutions recognize token claims, unless regulatory and legal frameworks adapt, and unless farmers possess the conversion capacities (education, digital access, literacy, trust), these mechanisms remain promising but fragile. On the first sub-question, what benefits and mechanisms? Tokenization enables new forms of collateral, more direct value chain payment structures, and increased transparency. On the second research question, barriers, are many: infrastructural, regulatory, institutional, literacy, and social trust barriers. On the third sub-question concerning influence over power and trust: tokenization shifts some power toward farmers, e.g., enabling them to prove yields or demand payment; however, unequal power remains, especially where platform owners, lenders, or regulators retain control over rules, data access, or adjudication. Trust increases where token claims are transparent and auditable, but autonomy concerns persist.

6.3 Limitations of the Study

This study has several limitations. It is case-based and focused solely on Zimbabwe; limiting generalizability, conditions may differ in other countries or regions. The relatively small sample (32 participants) was diverse in stakeholder type but may not include all possible voices, such as informal intermediaries, traditional leaders, or very remote smallholders. Besides, responses can be subject to social desirability bias: interviewees can overstate benefits of new technology because it is new or because interviewers have a background in working with innovation discourse. Even the risk exists that some of the interviewees had limited technical knowledge of tokenization and this might have biased their responses. These limitations impact the inferences made from the data. The single-country case study means that findings are context-bound, while the potential for social desirability bias can result in overstatement of technology benefits. Limited technical understanding among some farmer participants implies underreporting of challenges associated with usability and design issues. It follows that while this study identifies key barriers and potentials in a robust way, the relative weight of such factors and their general applicability need further testing in comparative and quantitative studies.

6.4 Directions for Future Research

Future research can quantitatively analyze the earning, productivity, or risk-reduction effects of tokenization for smallholders, and whether benefits felt at a qualitative level are captured in material economic outcomes. Comparative case studies in different African nations can analyze how different regulatory, infrastructural, and institutional factors affect tokenization success. Longitudinal studies would investigate how trust, power dynamics, legal status, and capability development evolve over time after adoption. Secondly, action research or pilot interventions co-designed with the farmers may illustrate which design choices (e.g., user experience, data protection, backup modes) make tokenization accessible. Finally, gendered and youth-based differential effect studies may reveal whether tokenization increases or reduces inequalities between smallholders.

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8. CONFLICTS OF INTEREST STATEMENT

The authors declare that they have no conflicting interests in publishing this paper. All the authors have made disclosure of any potential conflict of interest, including financial, personal, or professional relationship that might be perceived to influence the research that has been submitted for publication in this manuscript.

9. AI DECLARATION

The researchers declare that this manuscript represents the collaborative intellectual efforts and original contributions of authors. The authors employed QUILBOT grammar check to improve the grammar and readability of the paper. Also, the authors declare that no artificial intelligence (AI) tools were used to generate this manuscript; QUILBOT was only used to improve the grammar and readability of the paper.

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